The documentation and process conversion measures necessary to comply with this revision shall be completed by 18 November 2005 and the requirements of paragraph 4.3.2 shall be completed by 10 August 2006.

INCH-POUND
MIL-PRF-19500/116M
10 August 2005
SUPERSEDING
MIL-PRF-19500/116L
13 July 2001

* PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON, SWITCHING, TYPES 1N4148-1, 1N4148UR-1, 1N4148UB, 1N4148UB2, 1N4148UB2R, 1N4148UBCA, 1N4148UBCC, 1N4148UBD, 1N914, 1N914UR, 1N4531, AND 1N4531UR, JAN, JANTX, JANTXV, JANHC, AND JANKC

JANS1N4148-1 (see 6.4). Device types 1N914 and 1N4531 are inactive for new design.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

* The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

* 1.1 Scope. This specification covers the performance requirements for switching diodes. Three levels of product assurance are provided for each device type as specified in MIL-PRF-19500. Two levels of product assurance are provided for each unencapsulated device type.

* 1.2 Physical dimensions. See figures 1 (axial leads), 2 (DO-213AA), 3 (UB), 4 (UB2), and 5 (die).

* 1.3 Maximum ratings. Unless otherwise specified T_A = +25°C.

Type	V_{BR}	V_{RWM}	$I_{O(PCB)}$ T_A = 75°C (1) (2)	I_{FSM} t_p = 8.3ms	T_{STG} & T_J	$R_{\Theta JL}$ L = .375 inch (9.53 mm) (1)	$R_{\Theta JEC}$ (UR) (1)	$R_{\Theta JSP}$ (UB) (1) (3)	$R_{\Theta JA}$ (2)
	V dc	V (pk)	mA	A (pk)	°C	°C/W	°C/W	°C/W	°C/W
1N4148-1 1N4148UR-1	100	75	200	2	-65 to +175	250			325
							100		
1N4148UB, 1N4148UB2, 1N4148UB2R, 1N4148UBCA, 1N4148UBCC, 1N4148UBD					-65 to +200			120	325
1N4531 1N4531UR					-65 to +175	250			325
							100		
1N914 1N914UR						250			325
							100		

(1) For temperature-current derating curves, see figure 6.

(2) See figures 7, 8, and 9 for thermal impedance curves. T_A = +75°C for both axial and Metal Electrical Face (MELF) (UR) on printed circuit board (PCB), PCB = FR4 - .0625 inch (1.59 mm) 1-layer 1-Oz Cu, horizontal, in still air; pads for (UR) = .061 inch (1.55 mm) x.105 inch (2.67 mm); pads for axial = .092 inch (2.34 mm) diameter, strip = .030 inch (0.76 mm) x 1 inch (25.4 mm) long, lead length L ≤ 0.187 inch (≤ 4.75 mm); $R_{\Theta JA}$ with a defined PCB thermal resistance condition included, is measured at I_O = 200 mA dc.

(3) $R_{\Theta JSP}$ refers to thermal resistance from junction to the solder pads of the UB package.

* Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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1.4 Primary electrical characteristics at T_A = +25°C, unless otherwise indicated.

Type (1)	V_{F1}		V_{F2}		I_{R1} at V_R = 20 V dc	I_{R2} at V_R = 75 V dc
	I_F mA dc	V dc	I_F mA dc	V dc	nA dc	nA dc
1N4148-1	10	0.8	100	1.2	25	500
1N914	10	0.8	50	1.2	25	500
1N4531	10	0.8	100	1.2	25	500

Type (1)	I_{R3} at V_R = 20 V dc T_A = 150°C	I_{R4} at V_R = 75 V dc T_A = 150°C	t_{fr} at V_{fr} = 5.0 V dc (pk) and I_F = 50 mA dc	t_{rr}
	µA dc	µA dc	ns	ns
1N4148-1	35	75	20	5
1N914	35	75	20	5
1N4531	35	75	20	5

(1) Primary electrical characteristics for surface mount devices are equivalent to the corresponding non-surface mount devices unless otherwise noted.

2. APPLICABLE DOCUMENTS

* 2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

* 2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

* DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

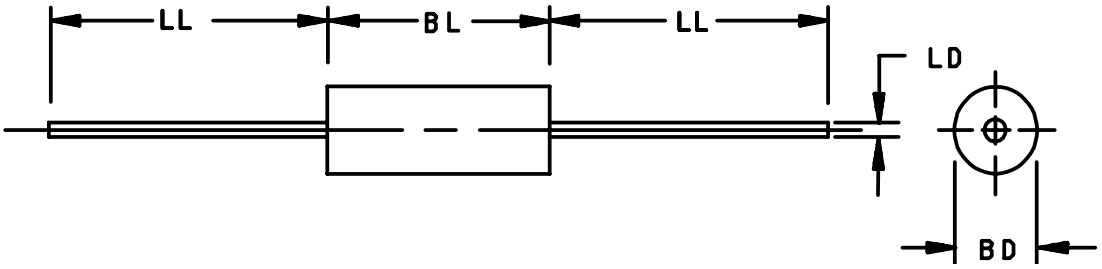
* DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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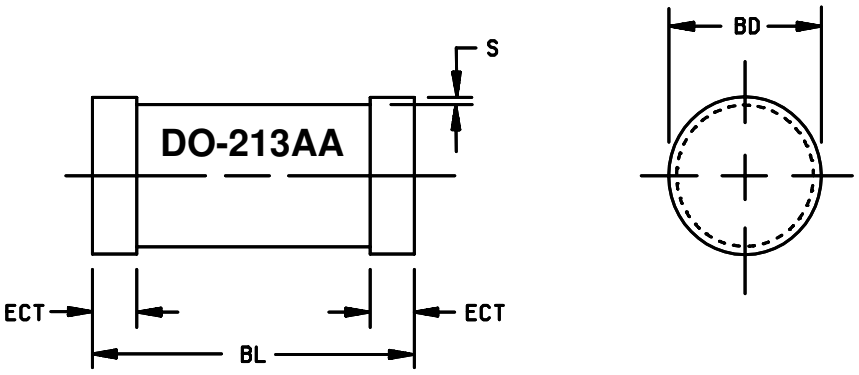
Types	Dimensions				
	Ltr	Inches		Millimeters	
		Min	Max	Min	Max
1N4148-1 1N914 (DO-35)	BD	.056	.075	1.42	1.90
	BL	.140	.180	3.56	4.57
	LD	.018	.022	0.46	0.56
	LL	1.000	1.500	25.40	38.10
1N4531 (DO-34)	BD	.050	.075	1.27	1.90
	BL	.080	.120	2.03	3.05
	LD	.018	.022	0.46	0.56
	LL	1.000	1.500	25.40	38.10

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. In accordance with ASME Y14.5M, diameters are equivalent to Φx symbology.

Types 1N4148-1, 1N914, and 1N4531.

* FIGURE 1. Physical dimensions.

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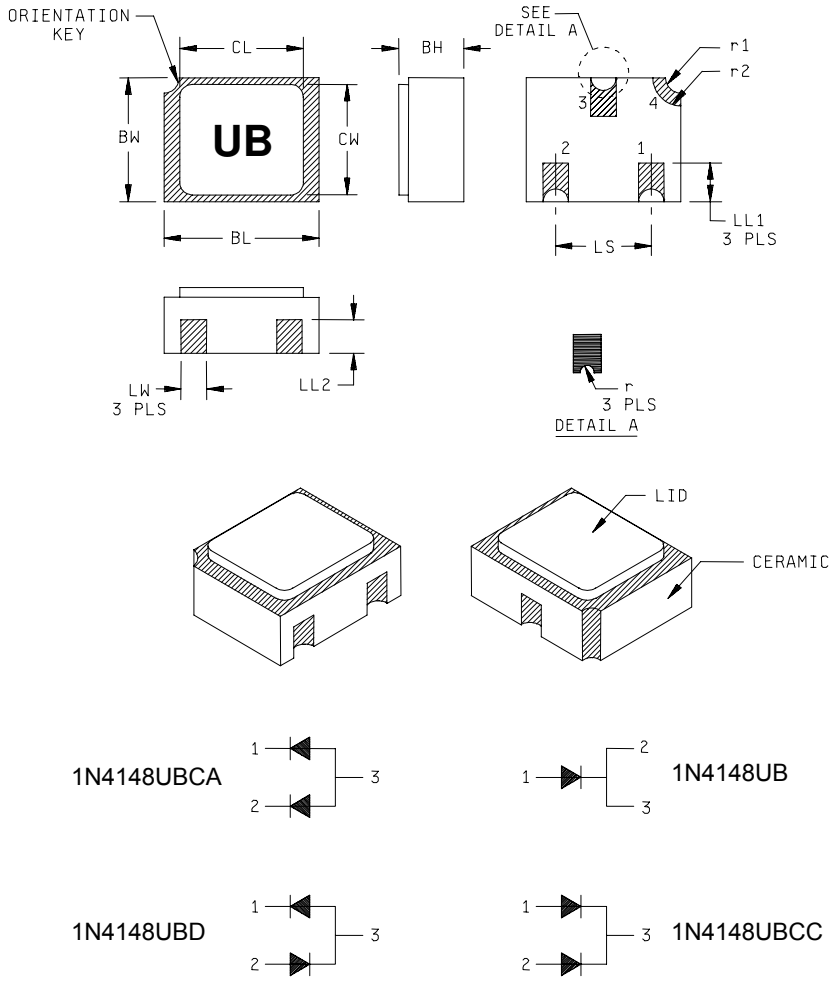
Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BD	.063	.067	1.60	1.70
BL	.130	.146	3.30	3.70
ECT	.016	.022	0.41	0.55
S	.001 min		0.03 min	

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Dimensions are pre-solder dip.
4. Referencing to dimension S, minimum clearance of glass body to mounting surface on all orientations.
5. In accordance with ASME Y14.5M, diameters are equivalent to Φx symbology.

Types 1N914UR, 1N4148UR-1, and 1N4531UR.

FIGURE 2. Physical dimensions (DO-213AA).

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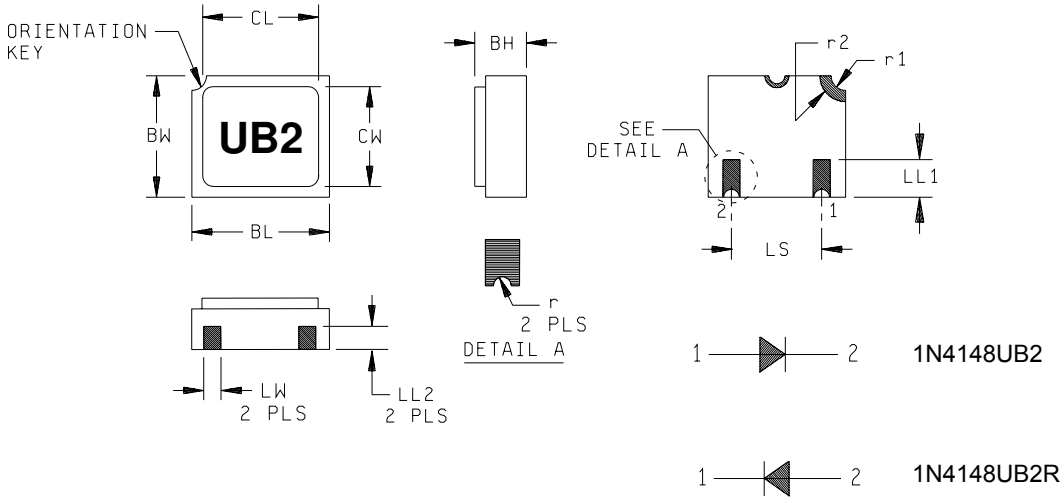


Symbol	Dimensions				Symbol	Dimensions			
	Inches		Millimeters			Inches		Millimeters	
	Min	Max	Min	Max		Min	Max	Min	Max
BH	.046	.056	0.97	1.42	LL2	.017	.035	0.43	0.89
BL	.115	.128	2.82	3.25	LS	.071	.079	1.81	2.01
BW	.085	.108	2.41	2.74	LW	.016	.024	0.41	0.61
CL		.128		3.25	r		.008		0.20
CW		.108		2.74	r1		.012		0.31
LL1	.022	.038	0.56	0.96	r2		.022		0.56

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Ceramic package only.
3. Hatched areas on package denote metallized areas. Pad 4 = Shielding, connected to the lid.
4. In accordance with ASME Y14.5M, diameters are equivalent to Φx symbology.

* FIGURE 3. Physical dimensions, surface mount (UB version).

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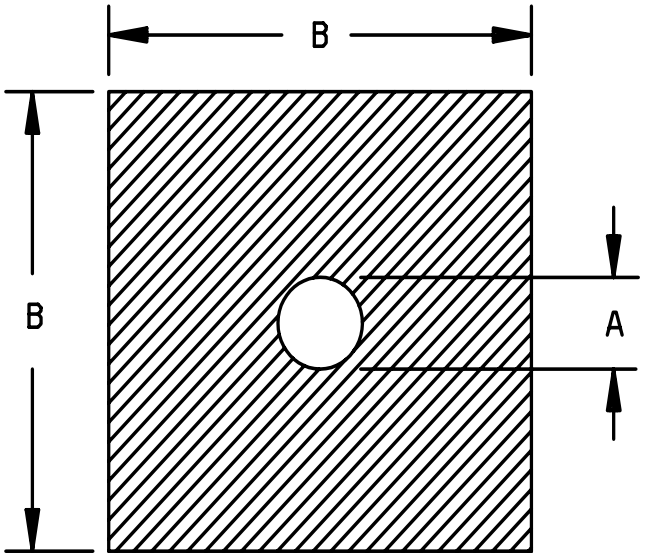


Symbol	Dimensions				Symbol	Dimensions			
	Inches		Millimeters			Inches		Millimeters	
	Min	Max	Min	Max		Min	Max	Min	Max
BH	.046	.056	0.97	1.42	LL2	.017	.035	0.43	0.89
BL	.115	.128	2.82	3.25	LS	.071	.079	1.81	2.01
BW	.085	.108	2.41	2.74	LW	.016	.024	0.41	0.61
CL		.128		3.25	r		.008		0.20
CW		.108		2.74	r1		.012		0.305
LL1	.022	.038	0.56	0.96	r2		.022		0.559

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Ceramic package only.
3. Hatched areas on package denote metallized areas. Pad 4 = Shielding, connected to the lid.
4. In accordance with ASME Y14.5M, diameters are equivalent to Φx symbology.

* FIGURE 4. Physical dimensions, surface mount (2 pin UB version).

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	Dimensions			
Ltr	Inches		Millimeters	
	Min	Max	Min	Max
A	.0059	.0061	.150	.155
B	.0130	.0170	.330	.430

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Element evaluation accomplished utilizing TO-5 package.
4. The physical characteristics of the die are:

Metallization:
Top (anode): Al
Back (cathode): Au
Al thickness: 25,000 Å minimum.
Gold thickness: 4,000 Å minimum.
Chip thickness: .010 inches (0.25 mm) ±.002 inches (0.05 mm).

FIGURE 5. Physical dimensions, JANHCA and JANKCA die.

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3. REQUIREMENTS

* 3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list (QML) before contract award (see 4.2 and 6.3).

* 3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

$R_{\Theta JBB}$ Thermal resistance junction to burn-in board.
SP Solder pad on UB devices.
UB Hermetic unleaded 3 terminal (LCC, Leadless Chip Carrier) package type.
UR Unleaded round package type designation.
V_{fr} Forward recovery voltage. Specified maximum forward voltage used to determine forward recovery time.

* 3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figures 1 (axial leads), 2 (DO-213AA), 3 (UB), 4 (UB2), and 5 (die).

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

* 3.4.2 Diode construction. All devices (except UB version) shall be metallurgically bonded, double plug construction in accordance with the requirements of MIL-PRF-19500. All glass diodes shall be designed with sufficient thermal compensation in the axial direction to optimize tensile and compressive stresses. Dimensional analysis is required of all materials used to achieve axial thermal compensation. Dimensional tolerances and corresponding coefficient of thermal expansion (CTE) shall be documented on the DSCC Design and construction Form 36D and shall be approved by the qualifying activity to maintain qualification. Dimensional tolerances shall be sufficiently tight enough to prevent excessive stresses due to the inherent CTE mismatch. The UB devices shall be eutectically mounted and wire bonded in a ceramic package. The UR version shall be structurally identical to the axial leaded versions except for end-cap lead attachment.

* 3.5 Marking. Marking shall be in accordance with MIL-PRF-19500. Manufacturer's identification and date code shall be marked on the devices. Initial container package marking shall be in accordance with MIL-PRF-19500. The polarity shall be indicated with a contrasting color band to denote the cathode end. The prefixes JAN, JANTX, and JANTXV can be abbreviated as J, JX, and JV, respectively. The part number may be reduced to J4148, JX4148, or JV4148. No color coding will be permitted for part numbering.

* 3.5.1 UR devices. For 'UR' version devices only, all marking, except polarity may be omitted from the body, but shall be retained on the initial container. Polarity marking of 'UR' devices shall consist as a minimum, a band or 3 contrasting dots around the periphery of the cathode.

* 3.5.2 UB devices. 'UB' devices do not require polarity marking.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

* 3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I herein.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

* 4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not require the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

* 4.3 Screening (JANTXV, JANTX, and JAN levels). Screening shall be in accordance with table IV of MIL-PRF-19500 and as specified herein. Specified electrical measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screening (see table IV of MIL-PRF-19500)	JANTXV and JANTX level
1a 1b	Not required Required (JANTXV only)
3a	Required
(1) 3c	Thermal impedance (see 4.3.3)
7a 7b	Not applicable (required for UB) Optional
9	Not required
10	Method 1038 of MIL-STD-750, condition A
(2) 11	I_{R1} and V_{F1}
12	See 4.3.2
(4) (4) 13	Subgroup 2 of table I herein; ΔI_{R1} = 100 percent of initial value or 15 nA dc, whichever is greater; ΔV_{F1} = 25 mV dc.
14a (5) 14b	Not applicable Required

(1) Thermal impedance shall be performed any time after sealing provided temperature cycling is performed in accordance with MIL-PRF-19500, screen 3 prior to this thermal test.
(2) Test within 24 hours after removal from test.
(3) When thermal impedance is performed prior to screen 13, it is not required to be repeated in screen 13.
(4) PDA ≤ 5 percent.
(5) For clear glass diodes, the hermetic seal (gross leak) test may be performed any time after temperature cycling.

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* 4.3.1 Screening (JANHC and JANKC). Screening of JANHC and JANKC die shall be in accordance with appendix G of MIL-PRF-19500. Burn-in duration for the JANKC level follows JANS requirements; the JANHC follows JANTX requirements.

* 4.3.1.1 JAN testing. JAN level product will have temperature cycling and thermal impedance testing performed in accordance with MIL-PRF-19500, JANTX level screening level requirements.

* 4.3.2 Power burn-in conditions. Power burn-in conditions are as follows (See 4.5.3): Method 1038 of MIL-STD-750, condition B. V_R = rated V_{RWM}; f = 50-60 Hz; I_O = 200 mA dc or I_F = 200 mA dc minimum. T_A = 75°C maximum. Adjust I_O or IF to achieve T_J = 125°C minimum or to the maximum current density limited by small die geometry. The maximum current density of small die shall be submitted to the qualifying activity for approval. Alternate mounting conditions shall be submitted to the qualifying activity for approval. With approval of the qualifying activity and preparing activity, alternate burn-in criteria (hours, bias conditions, T_J, and mounting conditions) may be used. A justification demonstrating equivalence is required. In addition, the manufacturing site's burn-in data and performance history will be essential criteria for burn-in modification approval.

* 4.3.3 Thermal impedance measurements. The thermal impedance measurements shall be performed in accordance with method 3101 or 4081 of MIL-STD-750, as applicable, using the guidelines in that method for determining I_H and I_M. t_{MD} shall be 70µs maximum, t_H shall be 10ms maximum. The thermal impedance limit shall comply with the thermal impedance graphs in figures 7, 8 and 9 (less than or equal to the curve value at the same t_H time) and shall be less than the process determined statistical maximum limit as outlined in method 3101 or 4081 of MIL-STD-750, as applicable. See group E, subgroup 4 of table II herein.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

* 4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table V of MIL-PRF-19500, table I herein, and as specified herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

* 4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500 and 4.4.2.1 herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2.

* 4.4.2.1 Group B inspection, table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500. Leaded samples from the same lot may be used in lieu of 'UR' suffix sample for life test.

Subgroup	Method	Conditions
B2	1056	0°C to +100°C, 10 cycles.
B2	1051	-55°C to +175°C, 45 cycles, including screening.
B2	2005	I_F = 100 mA, axial tensile stress = 8 lbs, T_A = +150°C; (not applicable to UR or UB package).
B3	1026	$V_{(pk)}$ = rated V_{RWM}; f = 50-60 Hz; I_O = 200 mA dc minimum; adjust T_A or I_O to obtain a minimum T_J of +150°C. (See 4.5.3)
B4	2101	Decap analysis; scribe and break only.
B6	1032	T_A = +175°C.

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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

* 4.4.3.1 Group C inspection, table VII of MIL-PRF-19500.

Subgroup	Method	Conditions
C2	1056	0°C to + 100°C, 10 cycles.
* C2	1051	-55°C to + 175°C, 45 cycles including screening.
C2	2036	Tension - test condition A; weight = 10 pounds, t = 15 s; lead fatigue = condition E (not applicable to 'UR' and 'UB' suffix types).
* C5	4081	L = .375 inch (9.53 mm), $R_{\Theta JL}$ = 250°C/W maximum; $R_{\Theta JEC}$ = 100°C/W; $R_{\Theta JSP}$ =120°C/W; (see 4.3.3), 22 devices, c = 0.
* C6	1026	1,000 hours minimum, $V_{(pk)}$ = rated V_{RWM}; f = 50 - 60 Hz; I_O = 200 mA dc minimum; adjust T_A or I_O to obtain a minimum T_J of +150°C. (See 4.5.3)

* 4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table IX of MIL-PRF-19500, and table II herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

* 4.5.1 Forward recovery voltage and time. Forward recovery shall be measured as the time interval between zero time and the point where the pulse has decreased to 110 percent of the steady-state value of V_F when I_F = 50 mA dc. The maximum rise time of the response detector shall be 1 ns.

4.5.2 Pulse measurements. Conditions for pulse measurements shall be as specified in section 4 of MIL-STD-750.

* 4.5.3 Free air power burn-in and life tests. The use of a current limiting or ballast resistor is permitted provided that each device under test still sees the full P_t (minimum) and that the minimum applied voltage, where applicable, is maintained through-out the burn-in period. Method 3100 of MIL-STD-750 shall be used to measure T_J.

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* TABLE I. Group A inspection.

Inspection 1/ 2/ 3/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal impedance	3101	See 4.3.3	$Z_{\Theta JX}$			°C/W
Forward voltage	4011	I_F = 10 mA dc (pulsed, see 4.5.2)	V_{F1}		0.8	V dc
Breakdown voltage	4021	I_R = 100 µA dc	V_{BR1}	100		V dc
Reverse current	4016	DC method, V_R = 20 V dc	I_{R1}		25	nA dc
Reverse current	4016	DC method, V_R = 75 V dc	I_{R2}		500	nA dc
Forward voltage	4011		V_{F2}			
1N914		I_F = 50 mA dc (pulsed, see 4.5.2)			1.2	V dc
1N4148-1, 1N4531		I_F = 100 mA dc (pulsed, see 4.5.2)			1.2	V dc
Subgroup 3						
High temperature operation:		T_A = +150°C				
Reverse current	4016	DC method, V_R = 20 V dc	I_{R3}		35	µA dc
Reverse current	4016	DC method, V_R = 75 V dc	I_{R4}		75	µA dc
Forward voltage	4011	I_F = 10 mA dc (pulsed, see 4.5.2)	V_{F3}		0.8	V dc
Low temperature operation:		T_A = -55°C				
Forward voltage	4011		V_{F4}			
1N914		I_F = 50 mA dc (pulsed, see 4.5.2)			1.3	V dc
1N4148-1, 1N4531		I_F = 100 mA dc (pulsed, see 4.5.2)			1.3	V dc
Subgroup 4						
Capacitance	4001	V_R = 0 V dc, f = 1 MHz, V_{sig} = 50 mV_{p-p} maximum.	C_1		4.0	pF
Capacitance	4001	V_R = 1.5 V dc, f = 1 MHz, V_{sig} = 50 mV_{p-p} maximum.	C_2		2.8	pF
Reverse recovery time	4031	Condition A, C ≥ 1 nF, I_F = I_R = 10 mA dc, R_L = 100Ω ±5 percent, $I_{R(REC)}$ = 1.0 mA dc, R ≥ 1000Ω.	t_{rr}		5	ns
Scope display evaluation	4023	Figure 4023-3, -7, -9, -10 only				
Subgroup 5						
Not applicable						

See footnotes at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/ 2/ 3/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 6						
Surge current	4066	Condition A (sine wave) $i_{f(surge)}$ = 2 A(pk), I_O = 0, V_{RM} = 0, 10 surges, 8.3 ms width each, one surge per minute, T_A = +25°C OR Condition B (square wave) $I_{F(surge)}$ = 4 A (pk), 10 surges, 1µs width each, duty factor = 0.0055%, T_A = 25°C				
Electrical measurements		See table I, subgroup 2				
Subgroup 7						
Forward recovery voltage and time	4026	I_F = 50 mA dc (see 4.5.1)	V_{fr} t_{fr}		5.0 20	V (pk) ns

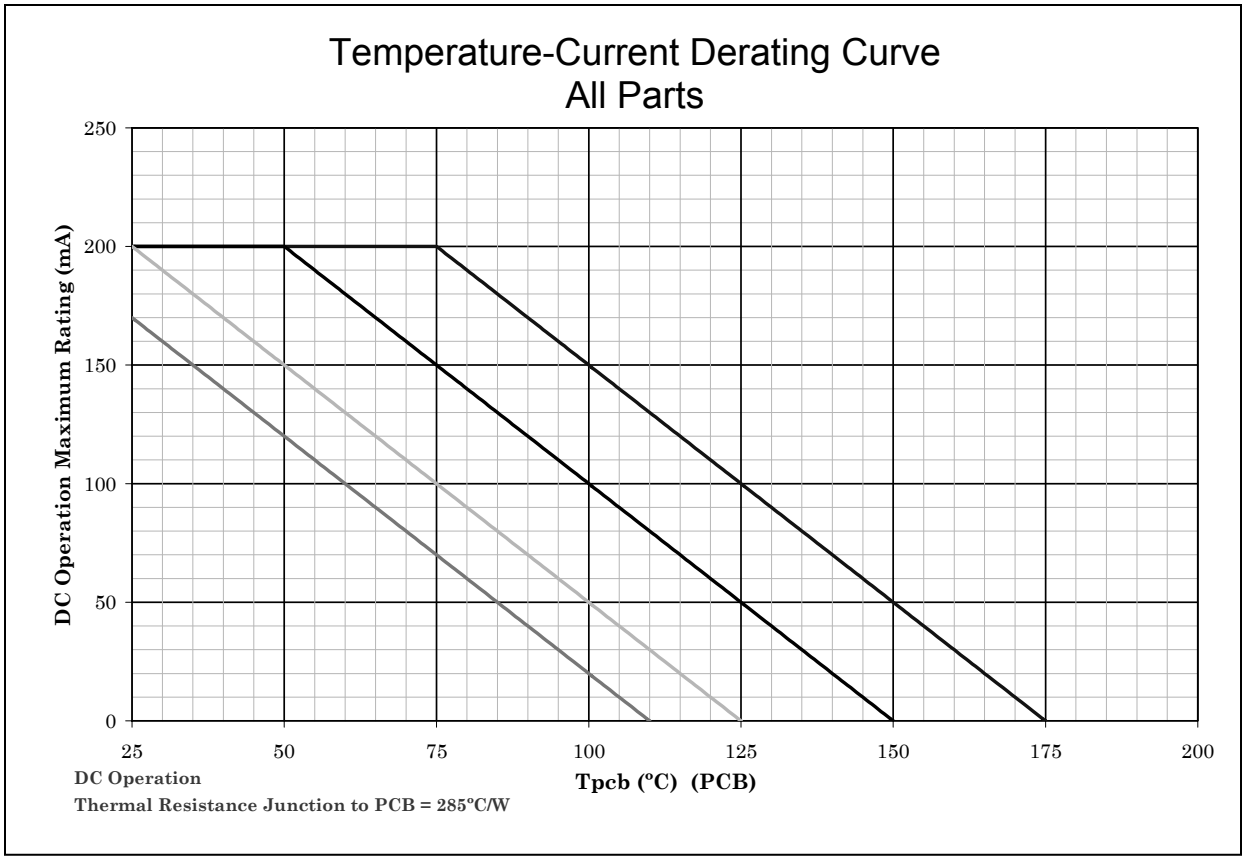
1/ For sampling plan, see MIL-PRF-19500.
2/ UBCA, UBCC, and UBD devices are to have each diode tested individually.
3/ Electrical characteristics for 'UB' and ' UR' suffix versions are identical to the corresponding no-suffix versions unless otherwise noted.

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* TABLE II. Group E inspection (all quality levels) for qualification and requalification only.

Inspection	MIL-STD-750		Qualification inspection
	Method	Conditions	
Subgroup 1			n = 45, c = 0
Thermal shock (glass strain)	1056	100 cycles 0°C to 100°C	
Temperature cycling	1051	500 cycles, -65°C to +175°C	
Hermetic seal	1071	Gross leak only. Fine and gross leak required for UB.	
Electrical measurement		See table I, subgroup 2.	
Subgroup 2			
Intermittent operating life	1037	10,000 cycles.	
Electrical measurements		See table I, subgroup 2.	
Subgroup 4			
Thermal impedance curves		A histogram of thermal impedance ($Z_{\Theta JX}$ and $R_{\Theta JX}$) is required on 116 devices using the supplier proposed optimal test conditions and thermal impedance limit. The approved thermal impedance conditions and limit for $Z_{\Theta JX}$ shall be used by the supplier in screening and for endpoint measurements as applicable. The approved thermal impedance conditions for $Z_{\Theta JX}$ shall be used by the supplier for all conformance inspection. Each supplier shall submit a thermal impedance ($Z_{\Theta JX}$) log-log plot using the best device, the worst device and the average device from the 116 devices histogram. In addition, the maximum thermal impedance limit curve from the slash sheet shall be included for reference to demonstrate that all curves conform. These four thermal impedance plots will all reside on a single log-log graph and will extend from DC steady state down to at least 1ms of heating pulse time.	
Subgroup 5			
Not applicable			
Subgroup 6			
ESD	1020		n = 3, c = 0
Subgroup 8			
Resistance to glass cracking	1057	Test condition B. Test until failure occurs or to a maximum of 25 cycles, whichever comes first.	n = 45
Subgroup 9			n = 22, c = 0
Monitored mission temperature cycling	1055	Not required for UB suffix devices.	
Electrical measurements		See table I, subgroup 2.	

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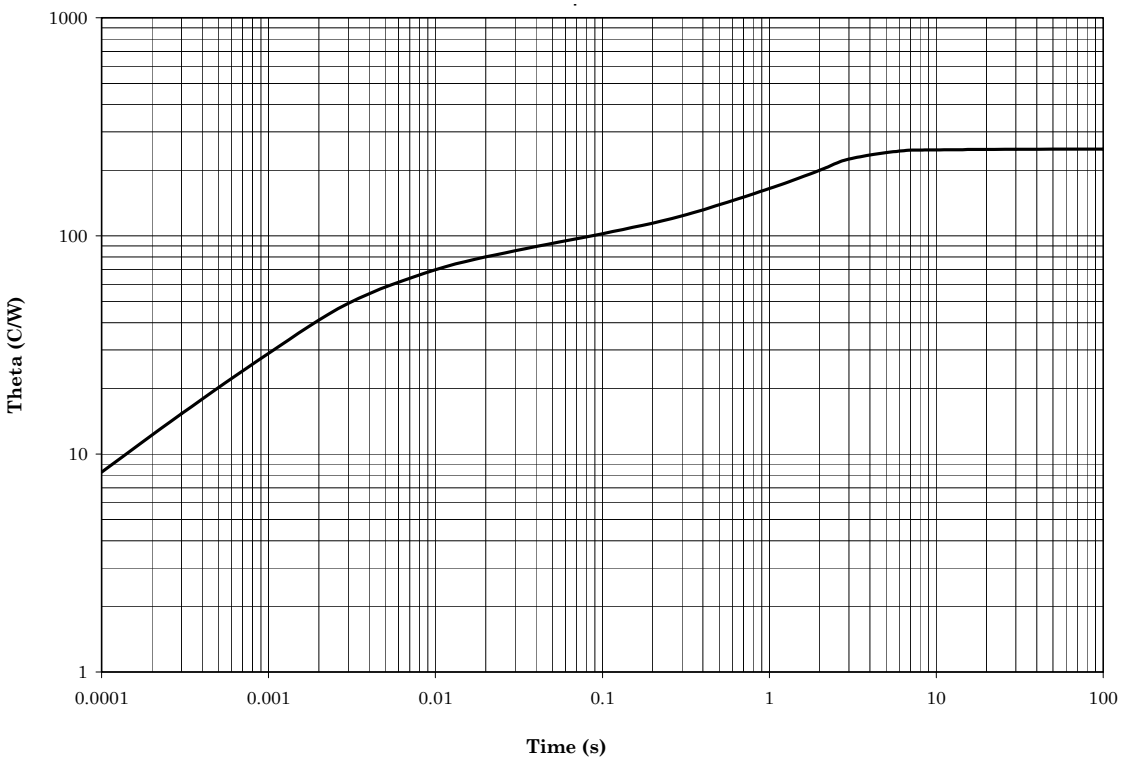
NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at $\leq T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq 175°C$) and current rating specified. (See 1.3.)
3. Derate design curve chosen at $T_J \leq 150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J \leq 125°C$, and 110°C to show current rating where most users want to limit T_J in their application.

* FIGURE 6. Temperature-current derating graph (all devices).

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1N4148-1, 1N4531, 1N914 DO-35 Axial T_L = 25°C
Maximum Thermal Impedance Plots

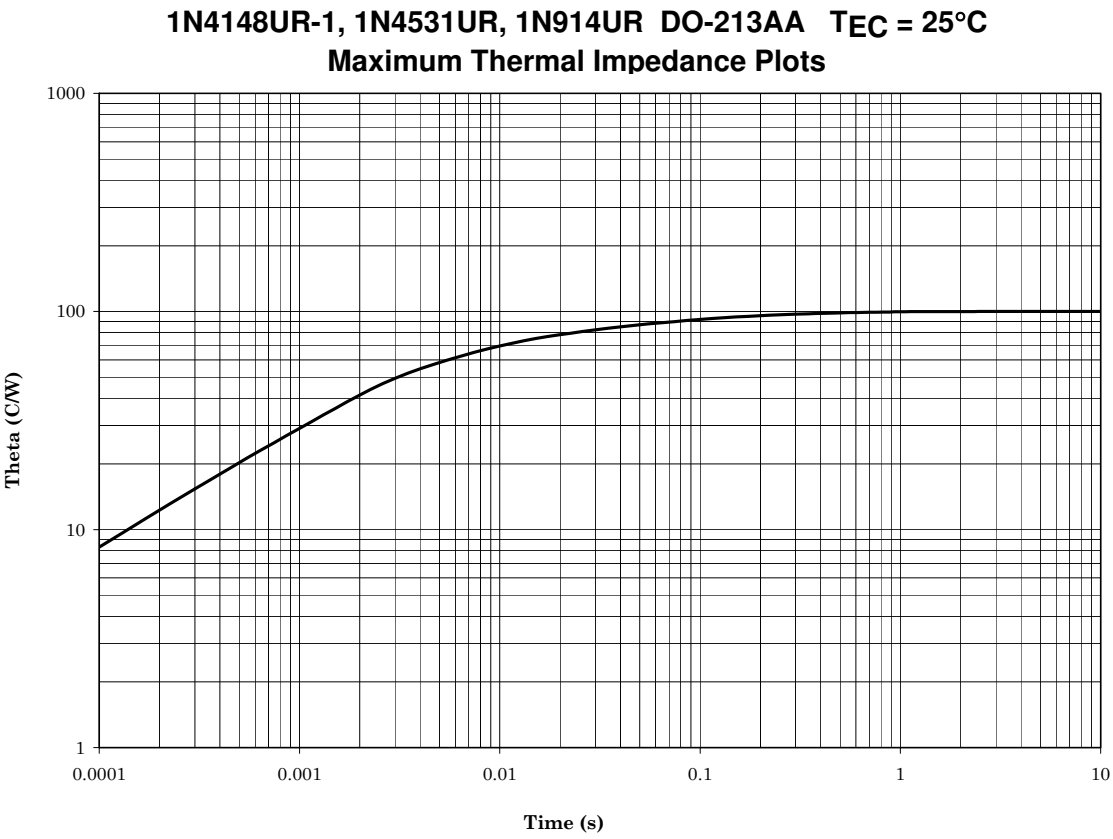


$R_{\Theta JL}$ = 250°C/W

1/ $Z_{\Theta JX}$ = 70°C/W maximum at t_H = 10ms.

* FIGURE 7. Thermal impedance (axial leads).

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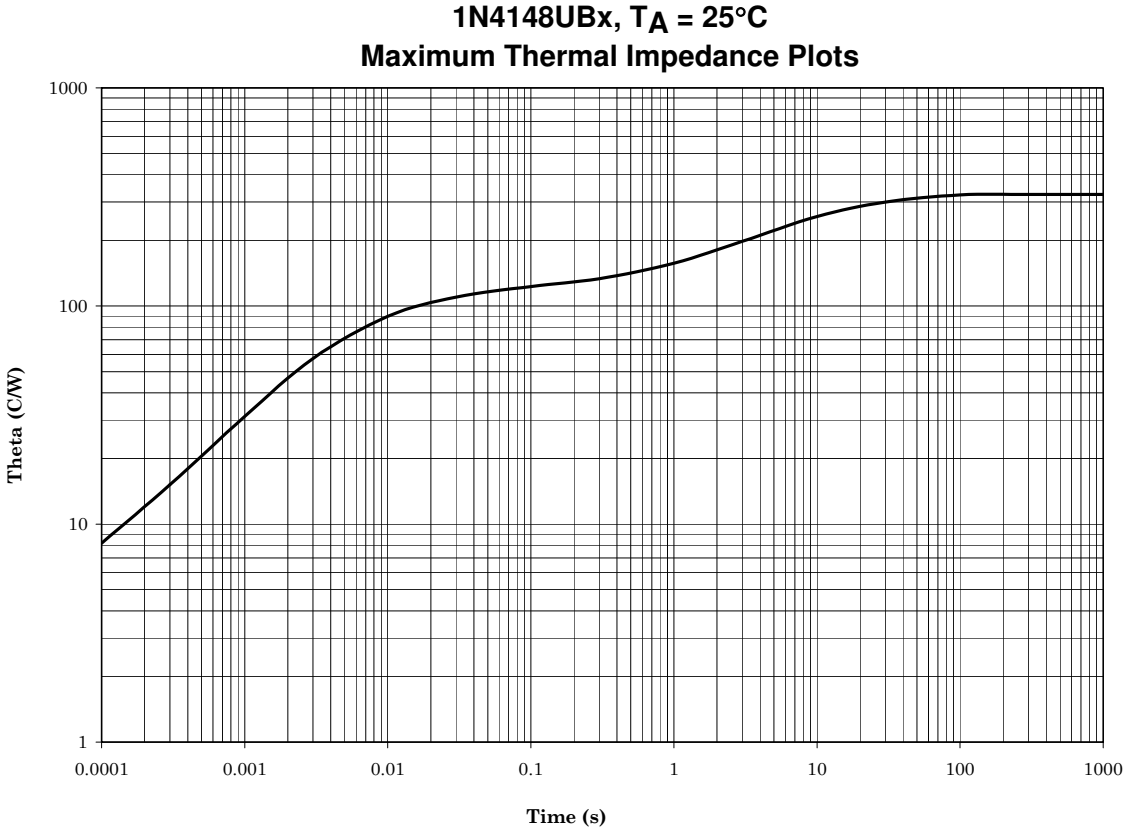


$R_{\Theta JEC}$ = 100°C/W

1/ $Z_{\Theta JX}$ = 70°C/W maximum at t_H = 10ms.

FIGURE 8. Thermal impedance (MELF surface mount).

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$R_{\Theta JA}$ = 325°C/W

1/ $Z_{\Theta JX}$ = 90°C/W maximum at t_H = 10ms.

FIGURE 9. Thermal impedance (UBx versions).

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5. PACKAGING

* 5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

* 6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

e. Destructive physical analysis when requested.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML-19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil.

* 6.4 Cross reference substitution list. JANS1N4148-1 is prohibited and will no longer be built or qualified. Devices in stock are acceptable provided the date code does not exceed 9208. A PIN for PIN replacement table follows, and these devices are directly interchangeable. The 1N4148 design is unsuitable for space flight applications. The JANS1N6642 will be used in place of the JANS1N4148-1. The 1N6638US, 1N6642US, and 1N6643US are directly substitutable for the 1N6638U, 1N6642U, and 1N6643U.

Non-preferred PIN	Preferred PIN
JANS1N4148-1 JANS1N4148UR-1 JANS1N6638U JANS1N6642U JANS1N6643U	JANS1N6642 JANS1N6642US JANS1N6638US JANS1N6642US JANS1N6643US

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* 6.5 Suppliers of die. The qualified die suppliers with the applicable letter version (e.g., JANHCA1N4148) will be identified on the QML.

JANC ordering information	
PIN	Manufacturer
	43611
1N4148	JANHCA1N4148, JANKCA1N4148

* 6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, MC
Air Force - 19

Preparing activity:
DLA - CC

(Project 5961-2734)

* NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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